



Agentic LLM Workflows for Personalized User Experience Questionnaire Generation

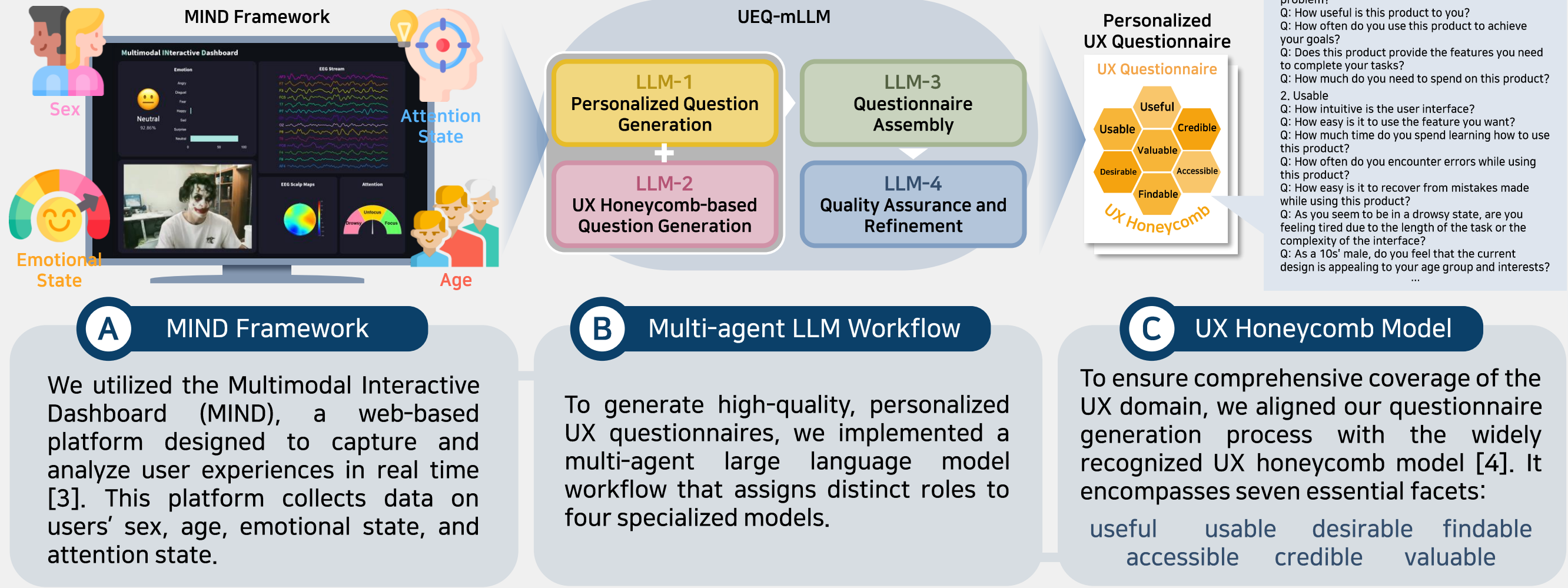
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Introduction

- The concept of personalized user experience (UX) evaluation has gained considerable traction in recent years, acknowledging the diverse needs, preferences, and contexts of users [1].
- Traditional questionnaires often encounter limitations such as generic questions, static formats, and a lack of personalization [2].
- This study introduces the **User Experience Questionnaire generation workflow using multiple LLMs (UEQ-mLLM)**, a novel approach that harnesses the capabilities of LLMs to create personalized and dynamic questionnaires.
- By integrating real-time user data with advanced language generation capabilities, **UEQ-mLLM** presents a promising solution for personalized and efficient UX assessment.

Methods



Experiments

- To simulate a diverse user population, we constructed a synthetic dataset comprising 80 hypothetical users. Participants were evenly distributed across sex (male, female) and age groups (teenagers, young adults, middle-aged adults, seniors). For each simulated user, we randomly generated emotional states (anger, disgust, fear, happy, sad, surprise, and neutral) and attention levels (focus, unfocus, and drowsy).
- To establish a benchmark for evaluating the performance of the **UEQ-mLLM**, we implemented a single LLM-based approach, denoted as **UEQ-sLLM**. This baseline model consolidated the functionalities of the four LLMs in the agentic workflow into a single LLM instance.
- To generate personalized UX questionnaires, we used Meta's open-source LLM, Meta-Llama-3.1-8B-Instruct [5].
- We adapted the G-Eval framework to evaluate the quality of the generated questionnaires. Four primary dimensions were established: **consistency, personalization, fluency, and relevance** [6].
- We utilized OpenAI's GPT-4 as the underlying LLM for G-Eval, configuring it with parameters $n=10$, $\text{temperature}=1$, $\text{top}_p=1$.

Results

- G-Eval scores were calculated for each questionnaire, with fluency rated on a 3-point scale and the remaining criteria on a 5-point scale.
- To facilitate comparison, scores were normalized to a 100-point scale.

Criteria	UEQ-mLLM	UEQ-sLLM	UEQ- mLLM vs. UEQ-sLLM	
			t	p
Consistency	4.08	2.08	20.23	< 0.001
Fluency	2.98	3.00	-2.61	0.010
Personalization	4.47	3.80	6.24	< 0.001
Relevance	4.48	4.04	5.85	< 0.001
Score	89.97	74.59	-	

The UEQ-mLLM consistently outperformed the UEQ-sLLM

Conclusions

- This study introduced **UEQ-mLLM**, a novel framework for generating personalized UX questionnaires.
- The proposed methodology offers a promising approach to enhancing UX evaluation by providing real-time insights into user behavior and preferences.

Acknowledgment

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References

[1] S. Valadi and I. D. Broneske, "Analysis of Current Usability and User Experience Questionnaires and Creating an Optimized Usability Questionnaire," Magdeburg: Universität Magdeburg, 2020.

[2] J. Morales, G. Rojas, and G. Cerda, "Towards a methodology to evaluate user experience with personalized questionnaires for the developments of custom systems," in International Conference on Human-Computer Interaction, 2022: Springer, pp. 267-281.

[3] H. Jo, J. Lee, H. W. Park, M. Kim, Y. Kim, and W. H. Lee, "Developing an Integrated Dashboard to Analyze Multimodal Data for User Experience Evaluation," in 2023 IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia), 2023: IEEE, pp. 1-4.

[4] P. Morville, "User experience design," Semantic Studios, vol. 21, 2004.

[5] A. Dubey et al., "The llama 3 herd of models," arXiv preprint arXiv:2407.21783, 2024.

[6] Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, and C. Zhu, "G-eval: Nlg evaluation using gpt-4 with better human alignment," arXiv preprint arXiv:2303.16634, 2023.

The UEQ-mLLM achieved a score of 89.97, representing a 20.62% improvement over the UEQ-sLLM, which has a score of 74.59.

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